

Debt Covenants and Corporate investments

HOW DOES FINANCING IMPACT INVESTMENT? THE ROLE OF DEBT COVENANTS
REPLICATION OF CHAVA AND ROBERTS (2008)

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1. Introduction

Does the way a firm borrows money affect how much it invests? This question matters because firms often use loans to pay for buildings, machinery and other long-term assets. A loan provides funding, but it also comes with rules that can limit the firm's decisions. These rules are called debt covenants. For example, a lender may require a firm to maintain a minimum current ratio or a minimum level of net worth. If the firm breaks a rule, the lender may gain the right to demand repayment or renegotiate the loan. This can happen even when the firm has made its loan payments on time.

A covenant violation may change the balance of power between managers and lenders. Before a violation, managers usually have more freedom to decide how to use the firm's resources. After a violation, lenders may demand tighter spending controls, reduce access to credit or limit new investment. This provides a possible explanation for why financing affects business investment. However, the difficult question is whether firms invest less because lenders gain control, or because their financial position was already becoming weaker.

Chava and Roberts (2008) investigate this question using US firm data from 1994 to 2005. They combine company accounts with loan-contract information. Because they know the actual covenant limits, they can identify when a firm crosses a limit and enters technical default. They find that investment falls by approximately one percentage point of beginning-of-quarter capital after a violation. Comparing observations close to the contractual limits strengthens their argument that the reduction reflects a change in creditor rights.

This study aims to replicate this paper. However, there is a data problem. The original study uses databases that cost money, such as Dealscan and Compustat, through a service called Wharton Research Data Services (WRDS). Many universities, including many in developing countries, do not pay for this service. This was checked and confirmed with the university librarian. The researchers in some countries therefore cannot check or update the result. For this study we use free public data.

The present study asks whether a similar negative relationship can be found using free public data for a later period. Its research question is: Are publicly disclosed debt-covenant violations associated with lower investment in the following quarter among the US firms studied during 2009 to 2019? This is an adapted replication. It follows the original investment measure and main panel regression approach, but changes the data sources, violation measure and observation period.

Four samples are analysed. Sample 1 matches an existing dataset of disclosed violations to financial information for 2009 to 2011. Samples 2 and 3 identify violations through searches of company reports filed with the US Securities and Exchange Commission (SEC). Sample 2 uses a 2012 to 2014 filing-search window, while Sample 3 extends the search through 2019. The combined sample merges Samples 1 and 3, removes duplicate observations and limits each source to its available coverage period. The largest primary regression sample contains 72 violating firms and 926 firm-quarter observations. One firm-quarter observation records information about one firm during one quarter.

Financial information comes from the SEC EDGAR XBRL Company Facts API. Investment is quarterly capital expenditure divided by net property, plant and equipment at the start of the quarter. The main regression examines whether a violation reported in the previous quarter is related to investment in the current quarter. Firm fixed effects account for lasting differences between firms, while time fixed effects account for changes shared across firms in the same period. Standard errors are clustered by firm because observations from the same firm may be related over time.

The study also examines whether accounting cutoffs can replace the unavailable contractual limits. These include a current ratio of 1.0 and a net-worth cutoff of zero. Further exercises use accounting measures as instruments for disclosed violations and combine liquidity pressure with high leverage. These analyses help assess the limits of the public-data approach, but they do not reproduce the original contractual-threshold design.

The main coefficients are statistically insignificant in all four samples. The combined coefficient is -0.0047, with a reported 95% confidence interval of approximately -0.045 to 0.036. Its estimated reduction is smaller than the original paper's main estimates, but the uncertainty is large. The interval includes both zero and a decline similar to the original finding. The study therefore does not establish a reliable investment reduction, but it also cannot rule out the original effect. Its contribution is to show what public disclosure data can reveal and where better information is needed.

2. Literature review

Why debt covenants may affect investment:

Agency theory explains why lenders place restrictions on borrowers. An agency problem arises when parties have different interests and one cannot fully observe or control the other's actions. Managers and shareholders may favour risky projects that lenders would prefer to avoid.

Shareholders can benefit if the project succeeds, while lenders may suffer if it fails and the firm cannot repay. Managers may also spend money on expansion that does not create enough value.

Chava and Roberts (2008) draw on the agency literature, including Jensen and Meckling (1976) and Smith and Warner (1979), to explain contractual restrictions. Covenants help lenders limit actions that could reduce the chance of repayment. They also set conditions under which lenders can intervene. A loan can therefore affect investment through the rights it gives creditors, as well as through the amount of funding it provides.

A second explanation concerns control rights. A loan contract cannot specify the correct response to every future event. Instead, it can define when lenders gain stronger rights to influence the borrower. The incomplete-contracting literature discussed by Chava and Roberts, including Aghion and Bolton (1992), provides a basis for this idea. When a firm complies with its covenants, managers retain more freedom. After a breach, lenders may gain greater bargaining power.

This change can lead to lower investment. A lender may require the borrower to preserve cash, repay debt or delay expansion. However, a contractual right is not always exercised in the same way. A lender may waive a breach or renegotiate the agreement without requiring a large spending cut. The theory gives a reason to expect lower investment, but the size of the response must be examined using data.

Dichev and Skinner (2002) show that violations occur fairly often and do not always indicate that a firm is close to bankruptcy. Breaking a covenant differs from missing a debt payment. A firm can continue paying its lenders while failing an accounting requirement. Studying these events helps examine contractual restrictions separately from severe financial distress.

2.2 What the original paper finds:

Chava and Roberts (2008) study firms with current-ratio or net-worth covenants. Their initial current-ratio sample contains 5,428 firm-quarter observations from 499 firms. Their initial net-worth sample contains 13,021 observations from 1,100 firms. These are sample-construction figures. The number of observations in each regression changes with the model, available variables and sample restrictions.

Their Table V reports negative violation coefficients between -0.015 and -0.007 in the full-sample models. Models using observations close to the covenant limits report coefficients

between -0.010 and -0.007. The paper summarises the decline as approximately one percentage point of capital per quarter, equivalent to roughly 13% of investment before the violation.

These units need careful interpretation. A coefficient of -0.010 means a fall of one percentage point in quarterly capital expenditure divided by beginning of quarter capital. It does not automatically mean spending falls by 1%. The percentage reduction in spending depends on the starting investment rate. This distinction matters when comparing the original estimates with the present study.

Table VII examines differences in the investment response using credit ratings, financial constraints, lending relationships, syndicate size and cash holdings. Larger declines among firms with greater agency or information problems support the creditor-control explanation. These are tests of differences in investment effects between groups. The present study does not directly repeat them because it does not construct the same measures and subgroup regressions.

2.3 Why the actual covenant limit matters:

Weak financial performance may cause both a violation and lower investment. Finding that these events occur together is therefore not enough to show that the violation caused the spending reduction. Chava and Roberts address this problem by examining observations close to actual contractual limits.

For example, consider a hypothetical loan requiring a current ratio of at least 1.5. A firm with a ratio of 1.49 has crossed the limit, while one with a ratio of 1.51 has not. Their accounting conditions are similar, but the lenders' contractual rights may differ. A sharp investment change at that boundary provides evidence about the role of those rights. This is the reasoning behind a regression discontinuity design.

Hahn, Todd and van der Klaauw (2001) explain why treatment assignment at the boundary matters. The cutoff must mark a change relevant to the event being studied. A researcher cannot obtain the same evidence simply by choosing a convenient accounting value. A current ratio of 1.0 shows whether current assets cover current liabilities, but does not reveal every firm's contractual limit.

A firm may violate a covenant while its ratio remains above 1.0. Another may have a ratio below 1.0 without breaking its loan agreement. A zero net-worth cutoff has the same problem. The contract may require a positive minimum or a limit that changes over time. The present

threshold exercises therefore study accounting conditions without establishing the contractual change needed for the original design.

2.4 Why the public disclosures are useful and what they miss:

Roberts and Sufi (2009) and Nini, Smith and Sufi (2012) provide a basis for studying violations through company disclosures. Their work connects creditor rights with financing and wider corporate decisions. SEC reports can therefore be useful when detailed commercial loan data are unavailable.

However, a reported breach and a breach calculated from contractual information are different measures. A disclosure may concern another covenant type, mention an earlier period or describe a breach already waived by the lender. The report's filing date may also differ from the breach date. These differences affect which events are observed and when they enter the regression.

Text searches can miss genuine violations if firms use unexpected wording. They can also find statements saying that no violation occurred. Checking the filing text and removing negated statements reduce these errors. Even after these checks, the absence of a detected disclosure does not prove that no breach occurred.

This matters for the combined sample. Sample 1 uses an existing disclosure dataset, while Sample 3 uses the researchers' own search procedure. Both identify reported breaches, but they may not capture the same events equally well. Limiting each source to its coverage period avoids coding uncovered periods as non-violation quarters. It does not make the two searches identical.

2.5 What the present paper can test:

The main hypothesis is that a disclosed violation is associated with lower investment in the following quarter, after allowing for the included controls and fixed effects. The null hypothesis is that this conditional relationship is zero. An insignificant result means there is not enough evidence to reject the null at the chosen significance level. It does not establish that the null is true.

The study also examines whether accounting cutoffs predict violations well enough to support its instrumental variable exercises. An instrument must be related to the violation measure. It must also satisfy other conditions, including affecting investment only through the violation in the proposed model. This requirement is difficult here because liquidity and leverage may

affect investment directly. Stock and Yogo (2005) also explain the problems caused by weak instruments.

Replication should be assessed using the direction and size of the coefficients, their uncertainty and the strength of the research design. Similar estimates alone do not establish successful replication. Different significance levels alone do not establish different effects.

3. Dataset description:

Chava and Roberts (2008) construct their analysis using quarterly firm-level financial data merged with loan-level data obtained from the Loan Pricing Corporation's Dealscan database. The database provides quarterly Compustat accounting data in addition to a precise record of dollar-denominated private lending agreements and their respective covenant terms. Critically, for the original study's identification strategy, Dealscan records the exact, contractually specified threshold for each loan. While the institution provides access to WRDS via the Refinitiv workspace, access to Dealscan and Compustat is restricted, and was thus unavailable for this empirical replication. The following section documents where our data construction necessarily diverges from the original study. We provide the explicit reasoning behind the alternative data construction and the resulting samples.

3.1 Treatment variable: Identifying covenant violations:

The treatment variable defined by Chava and Roberts (2008) is Bind, a covenant violation indicator specified by the following equation:

$$Bind_{it} = 1 \; if \; z_{it} - z_{it}^{0} < 0 \, , and \; 0 \; otherwise.$$

z_{it} denotes the firm's observed accounting variable in quarter t: The firms current ratio or net worth.

z_{it}^{0} denotes the covenant threshold: The minimum level (specified by the loan contract) of the accounting variable the firm must maintain.

The firm is considered to be in violation of its contract the instant the accounting variable falls below the specified covenant threshold, no matter how small the shortfall. A firm exceeding the threshold is considered "slack". The study's identification strategy rests on the following sharp rule: at the threshold, control rights are transferred to the lender, thus the distance from the threshold should not matter except through the violation itself. Chava and Roberts (2008) explicitly follow the data construction pipeline provided by Dichev and Skinner (2002). Dichev & Skinner (2002) emphasise that covenant violations (of the nature studied in this context) occur frequently, and are rarely followed by formal default. This frequency is precisely what enables a disclosure-based identification strategy, as it demonstrates that a meaningful number of real violation events are recorded in firms' own public filings, rather than remaining confined in private loan records.

The current study constructs the treatment variable using two independent sources.

3.1.1 Sample 1 (Disclosed violations, 2009-2011): The first sample was constructed using a publicly available dataset of SEC-filing-disclosed covenant violations (1996-2011). Sourced from Micheal Roberts' faculty research webpage (a co-author of the paper assigned for this replication), the dataset was originally compiled for Roberts & Sufi (2009) "Control Rights and Capital structure: An Empirical Investigation" Journal of Finance 64(4). The sample is disclosure-based, meaning violations are identified from firms' own quarterly and annual SEC filings. Since the SEC EDGAR's structured (XBRL) financial data provides consistent, reliable financial variables from 2009 onwards, this replication restricts Roberts and Sufi's (2009) sample to violations occurring from 2009 or later. This ensures the covenant data can be matched to the EDGAR-based financial variables used in our analysis.

3.1.2 Sample 2 and 3 (EDGAR full-text search): A second violation dataset was constructed by searching the full text of firms' SEC-10 and 10-Q filings using EDGAR's full text search system, following a similar approach to that of Roberts and Sufi (2009) in their construction of the preceding sample. The authors do not, however, explicitly document the phrases they employed in their search process. We searched for explicit covenant-violation language, with input phrases including (but not limited to): "in violation of certain financial covenants" "was not in compliance with certain financial covenants". Each candidate disclosure was verified on against the actual filing text to exclude negated statements (such as "we were NOT in violation of...") as well as on manual review to ensure exclusion of a category of false positives. Samples 2 and 3 are to be understood as applications of the same full-text identification procedure, differing primarily by the length of the search window. The obtained firm-quarter observations may extend beyond the nominal filing search window, as such, SEC filings may contain comparative information about previous reporting periods. Sample 2 uses a narrower 2012-2014 search window, while Sample 3 extends the search through 2019.

This full text approach aligns with the broader methodology applied by both Roberts and Sufi (2009) and Nini, Smith and Sufi (2012), who identify covenant violations directly from information contained in corporate filings. The approach enabled us to build samples extending beyond the 2011 period covered by the pre-existing disclosed-violation dataset, where XBRL coverage of small filers is more complete. The primary methodological distinction is between Sample 1 (which relies on a pre-existing dataset), and Samples 2 and 3 (which are self-constructed from the filing text).

Chava and Roberts (2008, Appendix B) discuss the challenges associated with precise measurement of covenant violations, even with direct access to the loan contracts. Thresholds may be dynamic, companies might choose to hold multiple overlapping loan packages (and these packages might have different covenant terms), and private lending agreements are often amended. Our data construction faces an analogous, measurement challenge. Using unstructured filing texts to identify, legitimate and correctly dated, disclosed covenant-violations. While the original study applies the structured loan-contract fields to filter and refine their data, this mechanism is not available to us. Thus, we address this challenge through the manual verification, as well as the negation-filtering processes, described in the preceding discussion.

3.2 Financial data SEC EDGAR:

Firm-level financial information (pertaining to capital expenditure, total assets, net property/plant/equipment, current assets and liabilities, net income, depreciation and debt) was extracted directly from SEC EDGAR's public XBRL Company Facts API, for every individual firm identified in the covenant-violation samples above. Capital expenditure is commonly reported on a fiscal-year-to-date basis in SEC EDGAR quarterly filings. To obtain true quarterly figures, we de-cumulated capital expenditures using each firm's actual reported fiscal year. This approach was taken over computing a calendar-year approximation, because we found that calendar-year approximation generated systematic mis-grouping for firms whose fiscal years did not coincide with the calendar year.

The extracted financial data panel was manually verified against the actual files 10-k/10-Q documents. The analysis revealed a duplication bug (where a small number of firms were extracted twice under different historical identifiers), as well as missing balance-sheet figures owing to inconsistent XBRL tag filings. Both problems were resolved, we deduplicated the firms appearing more than once, and where appropriate, the omitted balance-sheet figures were recovered using the accounting identities: Assets = Liabilities + Stockholder's Equity, along with Net PP&E = Gross PP&E - Accumulated Depreciation. Both figures are frequently tagged, even when the total is not, enabling the calculation. Manual verification also identified omitted submissions for some firm-quarters. This presents a structural limitation encountered when relying on publicly available EDGAR data for smaller reporting companies, particularly in the earlier segment of the sample period. The omitted data was submitted before the SEC's June 2011 universal XBRL mandate, and we confirmed this directly by searching the SEC site and observing the absence of an "Interactive Data" exhibit on the relevant filing.

3.3 Combined sample:

Construction of the combined sample was motivated based on statistical power. The objective was to obtain a sufficiently large sample that would enable adequate interpretation of the results obtained. The combined sample has more actual firms, not a different specification, threshold or control variable. Our approach aligns with the underlying statistical logic asserted by Hedges and Olkin (1985), who assert that pooling non-overlapping samples enhances precision.

Sample 1 (disclosed violations), built from the Roberts and Sufi (2009) pre-existing academic dataset and Sample 3 (full-text search) constructed from our own text search of SEC filings, were merged, using the following three step process:

(I) Prior to merging, we cross-checked firm identifiers (CIK) across both samples, finding only 6 of the 247 total firms appeared in both samples. This confirmed that the samples represent two legitimately different sets of companies, not the same firms counted twice.

(II) Each sample was then trimmed to its own verified window, prior to combining. This is the critical safeguard, preventing unobserved periods from being incorrectly coded as (Bind_it=0). Sample 1's violation data is only trustworthy through 2011 (where the source dataset's own coverage ends). Sample 3's violation data is only trustworthy through 2019 (where our own search window ends).

(III)The two trimmed data sets were subsequently concatenated. Following deduplication of the 6 identified overlapping firms.

The resulting sample is our largest, composed of 72 violating firms and 926 firm-quarters. The financial variables across all four samples are identically constructed, thus merging the samples merely alters the number of firms observed, leaving the definitions of any variables unchanged.

A limitation specific to the combined sample worth noting: Sample 1's violations were identified via the author's independent text phrase search, for which specific phrases were undocumented. Sample 3's violations were identified via our own, five specific text phrase each procedure, outlined in the preceding discussion. As such, the two processes, while similar regarding method, may diverge in implementation. Our phrase-based search built around explicit language such as: "in violation of certain financial covenants" may be more likely to uncover clearly worded disclosures, than whichever broader or narrower criteria applied by Roberts and Sufi (2009) in their construction of the dataset. The combined Bind variable should thus be understood as pooling two related, but perfectly homogenous measures of "disclosed

violation", constructed using independently- implemented versions of the same underlying identification approach, rather than as one single consistency defined treatment.

The following table provides a descriptive data summary.

Table 2: Descriptive data summary

	Chava & Roberts (2008)	Sample 1	Sample 2	Sample 3	Combined Sample
Violation Source	Loan Pricing Corporation's Dealscan database	Roberts and Sufi (2009) public dataset	Author-constructed (EDGAR full-text search)	Author-constructed (EDGAR full-text search)	Sample 1 + Sample 3 merged
Identification method	Threshold-based, according to accounting variable.	Pre-existing academic dataset. Roberts and Sufi's self-constructed filing text analysis.	5 keyword phrases. Manual negation and false-positive review.	The same 5 keyword phrases and manual negation and false-positive review, applied to an extended search window	Inherits 1+3
Violation search window	1994-2005	1996-2005	2012-2014	2012-2019	Firm specific: Each firm's own origin sample window.
Resulting firm-quarter coverage	1994-2005	2009-2011	2012-2014	2012-2019	2009-2019
Financial data source	CRSP-Compustat (quarterly merged)	SEC EDGAR XBRL Company Facts API	SEC EDGAR XBRL Company Facts API	SEC EDGAR XBRL Company Facts API	SEC EDGAR XBRL Company Facts API
Firms (violators)	~526 (≈ 185 by current ratio + ≈ 341 by net worth)	38	21	37	72
Firm-quarter observations.	6,256	133	243	802	926

Source: Author's own

3.4 Limitation regarding the treatment variable:

Across all samples, Bind strictly captures the publicly disclosed covenant violations, reported by firms in the observed 10-Q/10-K SEC filings, rather than observing all underlying breaches. Public disclosure can be selective, decisions to report are influenced by materiality, the degree of financial distress the firm faces, ongoing waiver negotiations and corporate reporting practices. These factors each may directly correlate with investment decisions. Thus, the obtained sample of disclosed violations may systematically diverge from the broader population of violations recorded in Dealscan. We acknowledge the potential selection bias as an additional empirical limitation, alongside our reliance on approximated covenant thresholds.

4. Variable definitions:

Variable definition adheres strictly with Chava and Roberts (2008, Appendix A), where the available data permits.

4.1 Control variables:

4.1.1 Cash flow:

Cash-flow is included to account for the impact of internal liquidity on financing constraints, in alignment with Cooper & Ejarque (2003), cited by the original study. Internally generated funds may ease investment constraints, independent of covenant dynamics.

4.1.2 Leverage:

Leverage serves a dual purpose, employed by Chava & Roberts as a proxy for the debt-overhang dynamics sanctioned by the work of Hennessy (2004), and as a direct balance sheet control.

Additional controls employed by Chava & Roberts such as Altman's Z-score, the Whited-Wu financial constraints index as well as abnormal accruals measures were not constructed for this study. These measures require market equity data or multi-year historical accounting detail beyond what we could access reliably and honestly, from SEC EDGAR's XBRL Company Facts API, particularly for the smaller, less closely followed, firms dominating our violator samples.

Table 3: Variable definitions.

Variable	Definition
Investment	Capital expenditure (De-cumulated to a true quarterly estimate) divided by lagged net property, land and equipment. This matches Chava & Roberts' Appendix A definition.
Bind (Treatment)	The indicator variable. Equal to 1 if the firm disclosed a covenant violation during the observed fiscal quarter, and 0 otherwise. Bind_t-1 is included, lagged by one quarter in alignment with Chava and Roberts' original specification.
Cash Flow	(Net income + Depreciation)/ Lagged net PP&E.
Log (Assets)	Natural logarithm of total assets.
Leverage	(Long-term debt + Short-term debt)/ Total assets.
Current Ratio	Current assets/Current Liabilities. Employed to construct the RDD running variable in the absence of the actual covenant threshold reported by Dealscan.
Distance	Current Ratio - 1.0. A constructed running variable. It is centred at a standard liquidity-distress threshold. Serves as a surrogate for the firm's true (unobserved) contractual covenant threshold.
Treat	An indicator variable. Equal to 1 if Distance <0 (where the current ratio lies below 1.0). This is a constructed proxy for the covenant-threshold crossing exploited in the original RDD.
Joint stress	An indicator variable. Equal to 1 if Treat = 1 and the firm's leverage lies above the sample median. This variable is applied to proxy for more severe agency and information issues, in alignment with Chava & Roberts section IV heterogeneity analysis.

Source: Author's own.

5. Empirical methodology:

5.1 Baseline panel specification:

The core specification directly follows Chava & Roberts equation 2:

$$\boldsymbol{Investment_{it}} = \boldsymbol{\alpha} + \boldsymbol{\beta} \cdot \boldsymbol{Bind_{i,t-1}} + \boldsymbol{\gamma} \cdot \boldsymbol{Controls_{i,t-1}} + \boldsymbol{\eta_i} + \boldsymbol{\gamma_t} + \boldsymbol{\varepsilon_{it}}$$

Where $\boldsymbol{\eta_i}$ captures the firm-specific fixed effects and $\boldsymbol{\gamma_t}$ captures the time fixed effects.

Chava & Roberts cite Peterson (2009), as the motivation behind their choice to cluster standard errors by firm, accounting for potential unobserved within-firm autocorrelation (which may otherwise superficially deflate standard errors) in panel settings of this nature. The current study adopts the same practice. Following the convention presented by Chava and Roberts, identifying $\boldsymbol{\beta}$, in the presence of firm-specific fixed effects, relies strictly on within-firm variation in Bind. A non-violating firm offers no identification information to $\boldsymbol{\beta}$, since its treatment statement is constant. The current study thus restricts the estimation sample to firms observed to have experienced at least one violation. This restriction is consistent with Chava & Roberts stated approach, who note that using the full sample (inclusive of non-violating firms) generates quantitatively similar results. The current study directly confirms this result using our own data in Section 6.1.

5.2 Proxy-threshold exercise motivated by Chava & Roberts' RDD:

Chava & Roberts directly attribute the methodological foundation of their study's design to Hahn, Todd and van der Klaauw (2001), defining their regression discontinuity approach as one wherein the probability of treatment changes discontinuously at a known threshold of an observed running variable. The threshold Chava & Roberts employ is precise and contractual: control rights are transferred to a firm's creditor immediately when the firm's covenant ratio crosses the definitive value asserted by its loan agreement, regardless of the magnitude of the breach. This enables Chava & Roberts to treat the distance to the defined threshold as locally uninformative for investment opportunities. Thus, a plausibly causal effect is isolated, as reported in their Table V.

The current study, owing to a lack of access to each firm's specific contractual threshold, is unable to replicate this design. The information required is provided exclusively on the Dealscan platform. As a surrogate, we constructed a standardised proxy threshold, namely, a

current ratio of 1.0. This measure is widely recognised as a liquidity-distress benchmark. Motivation behind the selection of this proxy is that current ratio covenants are one of the two covenant types analysed directly by Chava & Roberts, as evidenced by Table 1 of their paper. Our specification defines Treat and Distance as described in the preceding Section 4, and estimates:

$$\boldsymbol{Investment_{it}} = \boldsymbol{\alpha} + \boldsymbol{\beta} \cdot \boldsymbol{Treat_{it}} + \boldsymbol{\gamma} \cdot \boldsymbol{Distance_{it}} + \boldsymbol{\delta} \cdot (\boldsymbol{Treat_{it}} \times \boldsymbol{Distance_{it}}) + \boldsymbol{Controls} + \boldsymbol{\eta_i} + \boldsymbol{\gamma_t} + \boldsymbol{\varepsilon_{it}}$$

both on the full sample (inclusive of non-violators) as well as on a discontinuity sample restricted to $|Distance| < 0.20$, mirroring Chava & Roberts' ± 0.20 bandwidth, which they derive from Silverman's (1986) optimal window-width rule for their study's own current ratio and net worth samples.

Since Hahn, Todd and van der Klaauw's (2001) framework requires the explicit running variable's threshold to determine treatment assignment, we deliberately avoid defining the above estimation as a replication of Chava & Roberts RDD. This analysis is instead presented as an exploratory proxy-threshold exercise, motivated by Chava & Roberts design, distinct from the panel regression outline in the preceding 5.1 discussion, which represents this study's primary replication.

5.3 Fuzzy RDD (2SLS) and reduced-form:

In the aim of formally addressing the disparity between our constructed threshold and the true covenant threshold, our study additionally estimates a fuzzy RDD specification. The disclosed violation indicator (Bind) is instrumented with the constructed threshold-crossing indicator (Treat) by means of two-stage least squares.

6. Results:

The results presented below begin with the primary replication attempt, employing the same specification and identification logic Chava and Roberts report in Table V of the original study. Further, a proxy-threshold analysis is explored, motivated by the original paper's regression discontinuity design.

6.1 Primary replication: Panel regression across samples

The Chava and Roberts specification produced a negative coefficient on the lagged covenant-violation indicator, of approximately -0.015. As additional controls are adopted, the coefficient increases in magnitude, with estimates of approximately -0.010 to -0.015 (with the corresponding t-statistics consistently exceeding conventional significance thresholds). Our estimates are substantially smaller and are statistically insignificant across their specifications.

The results from four versions of the analogous panel regression are reported below. Three individually constructed samples as described in section 2, alongside the combined sample.

Table 4: Analogous panel regression results.

	Sample 1	Sample 2	Sample 3	Combined	Chava &Roberts (2008)
Bind_{t-1} coefficient	-0.0146	+0.0172	+0.0052	-0.0047	≈ -0.010 to -0.015
Standard error	0.0247	0.0159	0.0296	0.0207	-
p-value	0.556	0.278	0.860	0.821	<0.01
95% CI	(-0.063, 0.034)	(-0.014, 0.048)	(-0.054, 0.063)	(-0.045,0.036)	-
Firms (violators)	38	21	37	72	~ 500-1600
Firm-quarter observations	133	243	802	926	~ 4,900

Source: Author's own

The estimated coefficient on Bind is small in magnitude and not statistically distinguishable from zero at conventional significance levels across all four samples. The estimates further differ in sign across the individual samples. Sample 1 produces a negative coefficient of -0.0146, the closest estimate in magnitude to the original study's estimate. The estimate is however statistically indistinguishable from zero, with a considerably large standard error. We observe positive coefficients for Sample 2 and 3, completely diverging in direction from the original paper's estimate.

The combined sample is our largest (containing 72 violating firms along with 926 firm-quarter observations) and most precise specification, with the narrowest 95% confidence interval observed. Importantly, we see that doubling the sample size (relative to any individual sample) did not produce a point estimate closer to Chava and Roberts' reported magnitude. Rather, the coefficient moved further toward zero, and its confidence interval includes both modest negative and positive effects. This suggests that the discrepancy cannot be completely attributed to insufficient statistical power (owing to data limitations), even in our largest and most precisely constructed sample, the estimated effect is relatively small.

Thus, the combined specification does not provide evidence of the statistically significant investment reduction documented by Chava and Roberts.

This result is supported by the findings of Xu and Zhang (2024). Xu & Zhang found that, following violation of a bond covenant, it became significantly more difficult for the observed Chinese company to issue new bonds, which is consistent with the findings of studies on US companies. However, unlike the US studies, their findings reveal that the company's actual investment operations remained unchanged following the bond covenant violation. They attribute this divergence to differences in bondholder reactions, finding that bondholders vote to waive the violation rather than actually stepping in to exercise control. Thus, the violation, while flagged on paper, does not induce material change in terms of the party responsible for the company's investment decisions.

The confidence interval observed for the combined sample is sufficiently wide and suggests that the results should not be interpreted as proving that the covenant violations have no effect on investment. Specifically, the -0.045 to 0.036 95% confidence interval still permits an economically meaningful negative effect. As such, the appropriate conclusion our study draws is therefore that the available data do not provide statistically reliable evidence of the effect estimated by Chava and Roberts.

As a direct assessment of the sample-restriction logic in Section 4.1 (firms with fixed effects and no violation contribute no information to identifying β), our study re-estimates the combined sample-regression using the full set of 207 firms, inclusive of firms for which an identified violation had not been experienced. We observed a change in the coefficient from -0.0047 to -0.0066, leaving the essential conclusion unchanged. This provides empirical support to confirm the original study's stated approach, that restricting the primary estimation to violators does not drive our results and is thus not the reason why we received the near-zero result.

6.2 Exploratory proxy-threshold analysis: RDD and fuzzy RDD

Results from the proxy-threshold specification introduced distinct identification concerns. The current-ratio threshold of 1.0 serves as a provisional estimate for liquidity shortfalls, marking the point at which a firm's short-term assets no longer cover its short-term liabilities. It is not, however, representative of the firm's legally binding covenant threshold. Inconsistent signs were observed across samples and specifications, in addition to none of the coefficients reaching conventional significance.

Table 5: First stage Statistics and threshold sensitivity results.

Sample	F-stage F-statistic	Interpretation
Sample 1 (2009-2011)	3.06	Weak (Rule of thumb: F>10)
Sample 2 (2012-2014)	2.60	Weak (Rule of thumb: F>10)
Sample 3 (2010-2019)	0.77	Substantially Weak (Rule of thumb: F>10)
Combined Sample (2009-2019)	1.77	Weak (Rule of thumb: F>10)

Source: Author's own

Across the individual and combined samples, results from the first-stage regression of Bind on Treat suggest a weak relationship between the proxy and actual disclosed violations. The reported first-stage F-statistics were all observed to be materially below the conventional

Stock & Yogo (2005) strong-instrument threshold of 10. Further, the reduced form estimates of Investment on Treat produced statistically insignificant and small results for across all samples, echoing the main panel results.
These results establish this study's exploratory extension and are not to be interpreted as a direct replication of the original study's regression discontinuity design. The fuzzy-RDD estimates cannot be interpreted as a reliable causal estimate of the effect of covenant violations. The weak first-stage observed implies pronounced sensitivity of the 2SLS estimate to small amounts of variation in the treatment relationship.

The findings further emphasise that the current ratio (<1) is not an adequate empirical substitute to represent the firm's actual legally binding covenant threshold. Numerous firms with observed covenant violations remain above this constructed threshold, while many firms with current ratios below one are not observed to violate a covenant.

6.3 Net-worth based threshold analysis:
A second RDD proxy is computed, mirroring Chava and Roberts' examination of two covenant types throughout their study, as reported in Table I Appendix A of the original study's paper. The second RDD proxy estimated is a net-worth based threshold. Net worth is defined as: Net worth = assets – total debt, and is subsequently scaled by total assets, with zero representing the threshold. We estimate this using total debt, since full liabilities data-including non-debt items such as trade-payables, were not available from our EDGAR pull, a significant limitation.

Table 6: Net worth-based threshold analysis:

Sample:	Firm Quarters	Firms	Negative net worth (Treat_nw = 1)	Bind_lag=1 And Treat_nw=1 overlap	Regression result
Sample 2	263	27	3	0	Failed: Insufficient variation (Rank deficiency)
Sample 3	938	48	6	0	Failed: Insufficient variation (Rank deficiency)
Combined Sample	1,411	207	9	0	Failed: Insufficient variation (Rank deficiency)

Source: Author's own

The critical finding across the analysed samples indicates that in every single case, zero firm-quarters have both a disclosed violation and negative net worth simultaneously. Across the combined sample, negative net worth is observed for only 9 of the 1,411 firm-quarter, none coinciding with an actual, disclosed violation. Consequently, variation was insufficient to enable estimation of a co-efficient.

6.4 Threshold sensitivity analysis:

Six values of the current-ratio threshold are tested across three samples, motivated by Lee & Lemieux (2010) who explicitly recommend verifying whether results obtained are sensitive to the particular bandwidth or cutoff selected.

Table 7: Threshold sensitivity analysis

Threshold	Sample 2 p-value:	Sample 3 p-value:	Combined sample p-value:
0.75	0.57	0.27	0.26
0.90	**0.036**	0.75	0.34
1.00	0.36	0.84	0.98
1.10	0.60	0.72	1.0
1.25	0.68	0.17	0.35
1.50	0.27	0.51	0.45

Source: Author's own

The results observed were uniformly insignificant across the assessed threshold range, for sample 3 as well as the combined sample. A single isolated significant point sample is observed in sample 2 (at threshold = 0.90), which is interpreted as a false-positive, since this significance was not replicated on any other assessed sample or threshold.

6.5 Joint stress: Leverage-interacted heterogeneity

For deeper assessment, a second over-identified specification combining Distance with a leverage-interacted joint-stress indicator as two instruments for Bind is applied, motivated by the original paper. In Section IV, Chava and Roberts construct Table VII to test the heterogeneity hypothesis that the effect of a covenant violation on investment is non-uniform, relying directly on the severity of agency costs, information asymmetry as well as the firm's financial constraints. Their investigation interacts five proxies (credit rating, financial constraints index, lending relationships, syndicate size and cash holdings), finding that the investment decline after a violation, is only two to four times larger for firms facing more severe frictions.

The current study thus constructs a joint-stress measure to identify observations which are below the current ratio threshold and exceed the sample median in leverage. Four out of the five proxies used but Chava and Roberts require access to Dealscan's lender-side data to reproduce the original study's tests directly. We interact Distance with an indicator for above-median leverage, combined with Treat = 1 ("Joint stress"). This analysis tests the same underlying hypothesis, that the investment response is more concentrated among firms facing comparatively more significant financial constraints.

Results from the joint-first stage estimation (reported in table 7 below) indicate weak F-statistics for the joint first-stage estimation for every sample assessed. In addition, instrument agreement was not rejected in any sample according to the Sargan over identification test results, suggesting the two proxies are mutually consistent. The two proxies are however jointly not significantly stronger as in instrument for the true violation.

Table 8: leverage-interacted heterogeneity analysis

Sample	Joint F-statistic (Distance + joint_stress)	Joint F p-value	Sargan Statistic	Sargan p-value:
Sample 2 (2012-2014)	3.28	0.055	0.94	0.332
Sample 3 (2010-2019)	1.90	0.162	0.21	0.651
Combined Sample	3.41	0.035	0.79	0.375

Source: Author's own

7. Summary of results:

7.1 What the coefficients mean

The primary regressions produce different signs across the samples. Sample 1 has a coefficient of -0.0146, while Samples 2 and 3 have coefficients of 0.0172 and 0.0052. The combined coefficient is -0.0047. None is statistically significant at conventional levels. These results do not show a clear and reliably estimated direction for the relationship between disclosed violations and investment.

Sample 1 is close to some original estimates in magnitude and direction. However, its standard error is 0.0247 and its p-value is 0.556. The uncertainty is too large to conclude that this sample confirms a negative relationship. The positive estimates in Samples 2 and 3 also have high p-values, at 0.278 and 0.860. They do not establish that violations increase investment. A coefficient must be read together with its uncertainty.

The combined coefficient of -0.0047 means an estimated decline of 0.47 percentage points in the ratio of quarterly capital expenditure to lagged net property, plant and equipment. This describes the estimated relationship after allowing for the included controls and fixed effects. It does not establish that creditor intervention caused the decline.

For a simple illustration, if a firm's starting investment ratio were 8%, a reduction of 0.47 percentage points would bring it to 7.53%. This is only an example to explain the units, 8% is not a reported average for the present sample. The coefficient describes a change in the investment ratio, rather than a 0.47% fall in expenditure itself.

The combined standard error is 0.0207 and the p-value is 0.821. Under the fitted model and the null hypothesis of no relationship, the estimate is not unusual enough to reject the null. The p-value does not mean there is an 82.1% probability that violations have no effect. It tells us about the strength of the evidence against the zero relationship hypothesis under the assumptions of the test.

7.2 Why the confidence interval matters

The reported combined 95% confidence interval runs from approximately -0.045 to 0.036. In the investment-ratio units, it covers a decline of about 4.5 percentage points through an increase of about 3.6 percentage points. This wide range shows that the size and direction of the relationship are not estimated closely.

The interval includes zero, which agrees with the finding of statistical insignificance. It also includes -0.010, close to the original paper's main finding. A decline similar to the original result therefore remains compatible with the current estimate. The data do not distinguish that possibility clearly from no change or an increase.

This creates an important distinction. The study has not found strong evidence of a decline, but it has also not found strong evidence that there is no decline. To establish that an effect is very small, the study would need a sufficiently narrow interval around a small value or another suitable test. The current interval is too wide for that conclusion.

The smaller combined point estimate also does not prove that the effect has weakened over time. Comparing -0.0047 with the original coefficients describes a difference between estimates. Establishing a difference between the underlying effects would require an appropriate comparison that accounts for uncertainty and the different research designs.

7.3 What combining the samples achieves

The combined sample contains 72 violating firms and 926 firm-quarter observations. It is the largest primary regression sample. More observations can help, but they do not guarantee a more precise answer. The combined standard error of 0.0207 is smaller than those of Samples 1 and 3, but larger than Sample 2's standard error of 0.0159. The combined sample should therefore be described as the largest, rather than the most precise of all four.

Repeated observations from the same firm are not independent pieces of information. Investment decisions can remain related across quarters. Clustering standard errors by firm allows for this relationship. Precision also depends on changes in violation status within firms, variation in investment and the information in the other variables. Counting firm-quarter observations alone does not reveal how much information the regression has about violations.

The combined estimate remains near zero, but its interval still includes the original effects. Increasing the sample has therefore not removed the uncertainty. The result does not show that inadequate statistical power can be dismissed as an explanation. Combining the samples gives us more observations. However, the two sources may identify covenant violations differently because they use different search words and checking procedures. This may affect which violations are included in the combined sample.

7.4 What the full sample check tells us

Including firms without an identified violation changes the combined coefficient from -0.0047 to -0.0066. The draft reports that the main conclusion remains unchanged. This suggests that the restriction to violating firms is not the main reason for the insignificant result.

A firm whose violation status never changes does not provide its own before-and-after change in that status. However, these firms can still help estimate common time effects and the coefficients of other variables. They can therefore affect the final violation estimate indirectly. The full-sample check shows that the conclusion survives this sample change, rather than proving that the added firms have no role in estimation.

7.5 Why disclosure measurements can affect the results

The present study measures reported breaches, while the original measures whether accounting values cross actual contractual limits. These methods can identify different events. The disclosure sample may include covenant types beyond current-ratio and net-worth restrictions. It may also include breaches reported after a waiver or after negotiations with lenders.

Timing matters because the regression uses the previous quarter's violation status. If a breach is assigned to the reporting quarter instead of the quarter when it occurred, the regression may compare investment with the wrong event date. A missed disclosure can also leave a genuine breach coded as zero. These problems may weaken or otherwise change the estimated relationship. The reported findings do not establish the direction or size of any resulting bias.

The observed firms are also selected partly through the availability of disclosures and financial data. Their responses may differ from those of firms in the original loan samples. The present findings should therefore be applied to the studied samples with care. They do not give a general answer for every US borrower or every covenant breach.

7.6 What fixed effects can and cannot explain

Firm fixed effects account for characteristics that remain constant over time. For example, some firms may usually invest more because of their business type. Time fixed effects account for changes shared across firms in the same period. These features improve the comparison by allowing for persistent differences and common shocks.

However, they do not remove every changing factor within a firm. A fall in sales, the loss of a major customer or fewer profitable projects can affect both compliance and investment. The regression allows for the controls included in its model, but it cannot automatically separate all

these influences from lender intervention. Its coefficient is best described as a relationship after allowing for the included variables.

A waiver may also explain why some reported breaches are not followed by large spending cuts. This is a possible explanation, but the study does not measure waiver decisions or lenders' actions. It should not claim that waivers caused the weak findings. Claims that lenders became less strict during the later period would also need separate evidence.

7.7 Why the current-ratio instrument is weak

The current-ratio first-stage F-statistics are 3.06 for Sample 1, 2.60 for Sample 2, 0.77 for Sample 3 and 1.77 for the combined sample. The first stage checks whether the constructed threshold indicator helps explain the disclosed-violation variable. These low statistics indicate a weak relationship in the reported models.

This makes sense because a ratio of 1.0 is not every firm's contractual limit. The indicator separates firms whose current assets cover current liabilities from firms whose current assets do not. It does not necessarily separate borrowers who broke their loan contracts from borrowers who complied with them.

A weak first stage makes the instrumental-variable estimate unstable. The method uses investment changes related to the instrument to infer the relationship with violations. If the instrument explains very little variation in violations, small changes in the data can produce large changes in the final estimate. Conventional statistical tests may then be unreliable. The exercise cannot provide strong evidence of the causal effect of violations.

An F-statistic of 10 is a commonly used rough guide, rather than a universal critical value for every model. Instrument strength is also only one requirement. Low liquidity may directly restrict spending, so even a stronger first stage would not establish that the instrument affects investment only through violations.

7.8 What the net worth exercise show

The net-worth exercise has very few observations below its zero cutoff. In the combined diagnostic sample, only nine of 1,411 firm-quarter observations are below the cutoff, and none also has a lagged disclosed violation. The draft reports that the regression could not be estimated because of insufficient variation. This limits the conclusions available from that exercise.

The finding does not show that net-worth covenants are ineffective. A contract may require net worth well above zero. A borrower can therefore break that covenant while still having positive net worth. A zero cutoff may miss the events that matter under the contract.

The accounting definition also needs checking. Chava and Roberts define net worth as total assets minus total liabilities. The draft describes assets minus total debt. Liabilities can include trade payables and other obligations as well as interest-bearing debt. These definitions should not be treated as identical. Also, zero overlap alone does not establish the cause of a rank-deficiency error; that explanation must agree with the actual software output.

7.9 How to read the threshold sensitivity results

The sensitivity exercise considers six current-ratio cutoffs. Sample 2 has one p-value below 0.05, at a cutoff of 0.90, where the p-value is 0.036. The remaining results do not show a consistent significant pattern across samples and cutoffs. This provides little support for a stable result around the constructed boundaries.

When several tests are performed, an isolated significant result may occur by chance. That possibility calls for caution, but it does not prove that this finding is a false positive. The suitable conclusion is that the result is not robust across the reported checks. Focusing only on the significant cutoff would overstate the evidence. Changing the cutoff also does not reveal which value governs creditor rights. These checks show whether a finding depends on the researcher's chosen accounting value. They cannot replace the missing loan-contract information.

7.10 What the joint stress exercise can establish

Joint stress identifies observations with a current ratio below the constructed cutoff and leverage above the sample median. It represents a combination of liquidity pressure and high debt. The joint first-stage F-statistics range from 1.90 to 3.41, so the instruments remain weak in the reported models.

The Sargan p-values do not reject the imposed instrument restrictions. However, non-rejection does not prove validity. A test can fail to detect a problem, particularly with weak instruments or a small sample. Liquidity and leverage may also influence investment directly.

Using joint stress as an instrument does not itself estimate different investment responses for high-stress and low-stress firms. The exercise therefore does not directly repeat the original

paper's subgroup tests. It should be interpreted as an exploratory instrument exercise, without claiming that it confirms or rejects the original findings about differences between firms.

8. Comparison with original paper:

8.1 Features preserved in this replication

Both studies examine investment after violations using quarterly information. Both scale capital expenditure by capital at the start of the quarter and include a lagged violation indicator. The present study also uses firm and time fixed effects and firm-clustered standard errors. These choices preserve important parts of the original panel framework.

However, the present study does not construct every control or every measure used by Chava and Roberts (2008). Its firms and violation measure also differ. It is therefore an adapted replication of the main relationship, rather than an exact reproduction of the original analysis.

8.2 Differences in the estimated results

The original full-sample Table V coefficients range from -0.015 to -0.007, and its discontinuity coefficients range from -0.010 to -0.007. The current combined coefficient is -0.0047. The individual coefficients range from -0.0146 to 0.0172. The original study's consistently negative and statistically significant pattern is not recovered across the present samples.

Sample 1 is closest to the original estimates in magnitude and direction, but its uncertainty is much greater. The combined coefficient has the expected negative sign and a smaller estimated magnitude, but is also imprecise. Samples 2 and 3 have positive estimates without enough evidence to establish a positive relationship. These findings describe the estimates; they do not establish that the true investment response has changed.

The combined confidence interval includes the original paper's main negative coefficients. No formal comparison reported here establishes a difference between the underlying effects. A significant finding in the original paper and an insignificant finding here are not, by themselves, evidence of a significant difference between the studies.

8.3 Differences in data and event measurement

The original study uses loan-contract and accounting data for 1994 to 2005. The present study uses public disclosures and SEC financial information within coverage windows spanning 2009 to 2019. The original violation measure depends on actual current-ratio and net-worth

requirements. The present measure depends on whether a breach is detected in a company report.

These differences affect the observed events and their dates. A disclosure may concern another covenant type or a breach that has already been negotiated. A contract-based measure may identify a breach before a later report discusses it. Different findings are therefore possible even if the basic creditor-control mechanism remains relevant.

The later observation period may matter, but the current analysis cannot isolate its effect. The data source, firms, controls and violation measure change at the same time. The results do not establish that changes in lending practices caused the discrepancy. Separating this explanation from differences in measurement would require further evidence.

8.4 Differences in causal evidence

Chava and Roberts know the contractual boundary at which rights change. Comparing observations close to that boundary helps distinguish the change in creditor rights from small accounting differences. The present study does not have those boundaries. Its accounting cutoffs do not establish that creditor rights change at the selected values.

The weak first stages further limit the instrumental-variable exercises. They do not resolve the missing-threshold problem or provide reliable causal estimates. The leverage-based instrument exercise also differs from the original subgroup investment tests. The present study therefore provides less direct evidence about whether lender intervention causes investment to fall.

The comparison supports a careful conclusion. The public-data implementation does not recover the original pattern with comparable precision or causal support. At the same time, its uncertainty does not rule out a similar effect. This is an inconclusive replication of the investment finding, with useful evidence about the limitations of the chosen accounting proxies.

9. Conclusion:

This study examines whether publicly disclosed covenant violations are followed by lower investment among US firms observed within coverage windows spanning 2009 to 2019. It uses SEC financial information, an existing disclosure dataset and independently searched company reports. The main model follows important features of Chava and Roberts (2008), while adapting the analysis to public data.

The regressions do not establish a statistically significant investment reduction. The largest primary sample contains 72 violating firms and 926 firm-quarter observations and produces a coefficient of -0.0047. This represents an estimated decline of 0.47 percentage points in the investment ratio. However, the confidence interval is wide and includes both zero and the original negative estimates. The study therefore cannot confidently establish either that investment falls after violations or that violations have no meaningful effect.

The exploratory results show that general accounting cutoffs are weak substitutes for actual contractual limits. The current-ratio indicators predict disclosed violations poorly, the net-worth cutoff identifies very few observations, and the sensitivity checks do not produce a consistent pattern. Adding joint financial stress does not remove the weak-instrument problem. These exercises do not reproduce the original discontinuity design or its causal evidence.

The creditor-control explanation remains possible. Lenders may still affect investment after a breach, but the present data and methods do not identify that response precisely. Disclosure selection, event timing, sample size and differences in controls may contribute to the results. The study cannot determine how much each factor matters.

Future work should improve the dating and completeness of violation records, use a consistent search procedure and check the accounting definitions against the original paper. Actual covenant limits should be collected from publicly filed credit agreements where possible, including changes made during renegotiation. A larger sample would help, but better event measurement and a stronger research design are also needed. These changes would make it easier to assess whether a different estimate reflects a real change in investment behaviour or differences in how the studies measure and analyse violations.

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